

THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MECHANICAL ENGINEERING

MECH4412 Product Design and Development

Project Report

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1. Introduction

This project involves the construction of a semi-automatic ball shooter, either manual or auto control. The shooting mechanism should be automatic whereas the balls can be loaded manually. The shooter robot will be set in a fixed position in the game field and shoot three baskets within 5 minutes.

The target is a toy basketball hoop placed 1.5m away from the shooting zone, each hoop with different level of height. The team which finishes the three baskets with shortest time will be the champion. A map of the game field is shown below, along with the image of the target.

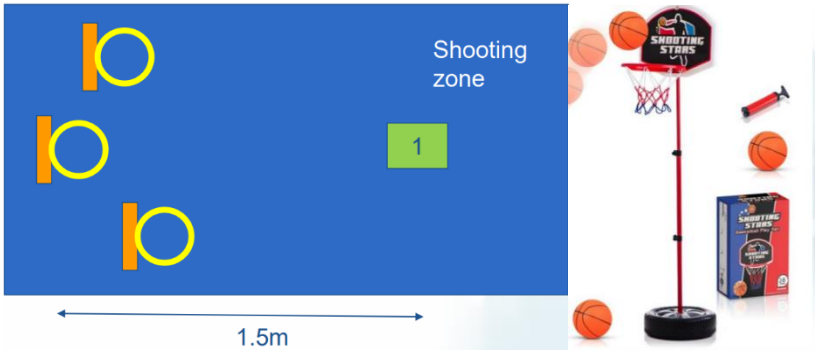


Figure 1 Shooting area and basketball hoop

There are some additional restrictions for the competition. First, the size of the robot is limited to 5 reams of A3 paper box, i.e., 42 × 29.7 × 25cm. Second, standard raw materials and components for designing the robot will be provided, such as motors (servo, stepper, dc), controller, microcontroller board, connection wire, plywood board, and aluminum profiles. An additional budget of HK $1500 will be allocated to each group for any additional materials.

2. Aims

The ultimate objective for the competition is to achieve the highest score by successfully shooting all three baskets within the shortest time. Prioritizing speed and efficiency in shooting are the critical winning tactics to attain the goal.

It is determined that the implementation of the counter-rotating wheel mechanism perfectly aligns with the project requirements, where it provides exceptional speed, consistency, and efficiency. To ensure precise and consistent angle adjustments, a combination of servo motors and stepper motors will be utilized. By implementing automation system, initial vertical and horizontal shooting angles can be set respective to each basket position and automatically reposition them for subsequent baskets. This approach can eliminate the need for manual angle adjustment, reducing potential time delays and enhancing overall accuracy.

3. Schedule

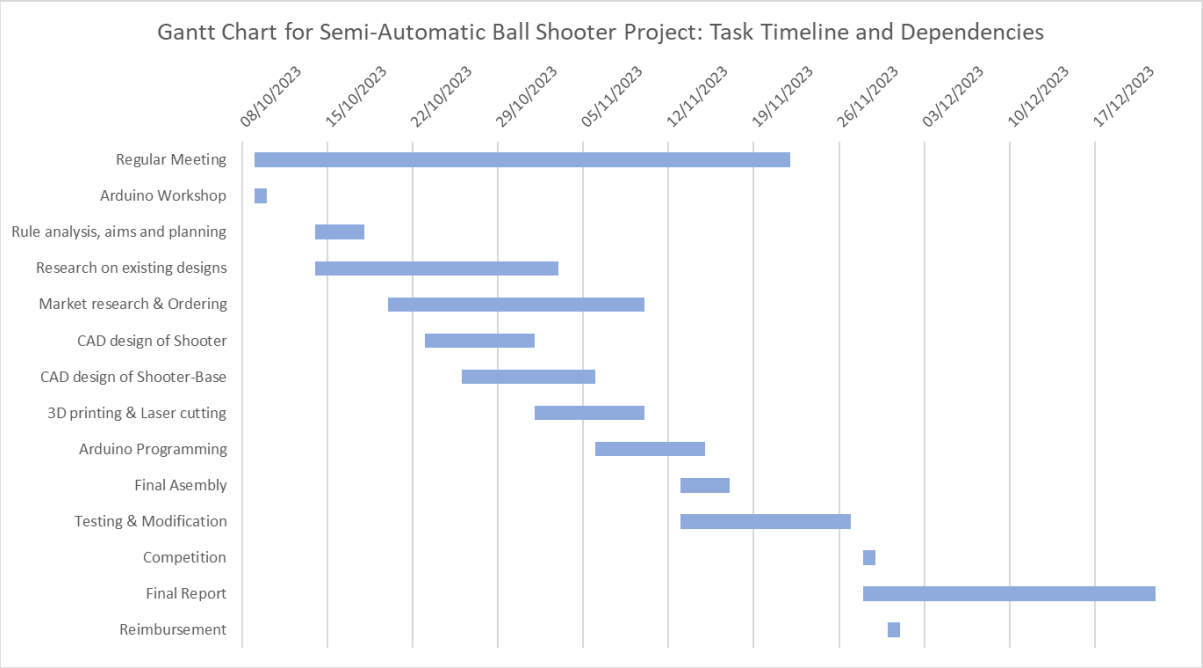


Figure 2 Task timeline

4. Duty Table

Name	**Role**
Chen Tien-Hao	SolidWorks parts modifications, construction of launcher
Hwang Donggyu	SolidWorks design, electronics, construction of launcher
Kim Daye	Electronics, construction of launcher
Kim Seonhyeok	SolidWorks design, electronics, construction of launcher
Xie Hanming	Procurement, construction of launcher

5. Research on Different Mechanisms

Robot Arm

The robot arm ensures consistent and precise shots, greatly enhancing our ability to make the shots in the shortest time. With its high precision and accuracy, the robot arm can repeatedly execute the same shooting motion, resulting in consistent and reliable performance and giving us a competitive edge by minimizing the variability in our shooting outcomes. Also, the programmability of the robot arm allows us to optimize the shooting technique. The arm emulating different throws and shots can be programmed, and the release point, angle, and force can be fine-tuned.

Despite its advantages, there are a couple of considerations when using a robot arm for our project. Firstly, there is the issue of cost, as acquiring, installing, and maintaining a robot arm can involve a significant initial investment. The advanced technology and precision required in a robot arm come with a higher price tag, which needs to be carefully evaluated against the potential benefits and financial feasibility. Additionally, the movement speed of a robot arm may impose limitations on shooting efficiency. While the arm can provide precise and accurate shots, its mechanical constraints may restrict its speed, which could potentially impact our ability to achieve a shorter completion time.



Figure 3 Robotic arm shooter

Pneumatic Piston

Pneumatic systems are known for their simplicity. They involve fewer moving parts

compared to other systems, making them more reliable and easier to maintain. This simplicity translates to a higher level of consistency in the shooting process, as there are fewer components that can potentially fail or require frequent adjustments. Pneumatic systems exclude the risk of overheating often happening on motors as well, making it more stable and reliable during our trials.

However, there are some drawbacks to consider when using a pneumatic system for shooting in our project. One notable limitation is the relative lack of precision compared to other systems. The variability in air pressure can introduce a degree of unpredictability into the shooting process, potentially resulting in less consistent ball speeds and trajectories. This variability can impact the accuracy and reliability of our shots, making it more challenging to consistently make the shots in the shortest time. Careful calibration and adjustment of the pneumatic system's pressure and release mechanisms will be needed.



Figure 4 Pneumatic piston shooter

Counter-rotating Wheels

Implementing a counter-rotating wheel system offers several advantages to ensure accurate and fast shooting of the hoops. One is the system's ability to adjust the speed and spin of the ball. By controlling the relative speeds of the counter-rotating wheels, not only the velocity of the ball but also the amount and direction of its spin can be modified. This adjustability allows us to fine-tune our shots, optimizing them for each specific hoop and increasing our chances of successfully hitting the targets in the shortest time. Additionally, the counter-rotating wheel system boasts a simple and durable design with fewer moving parts. This simplicity enhances the system's reliability, reducing the likelihood of breakdowns and

minimizing the need for frequent maintenance. The system's durability ensures that we can consistently perform at our best throughout the competition.

There are a couple of considerations when using a counter-rotating wheel system for our robot-shooting toy basketball competition. One potential issue is the wear and tear it can cause on the balls. The friction generated as the ball is propelled by the counter-rotating wheels can gradually degrade the ball's surface over time. Another challenge lies in achieving a high level of precision and consistency with the counter-rotating wheel system. Minor variations in the ball's position as it is fed into the machine, as well as wear on the wheels themselves, can introduce slight differences in the ball's trajectory and speed from shot to shot. However, these potential difficulties can be solved easily, and our solutions to the problems of wear and tear of the ball and the wheels as well as the variations in the ball’s initial position will be introduced in the latter (Final Design) part.



Figure 5 Counter-rotating wheels shooter

6. Design & Problem Solving

6.1 Initial Design

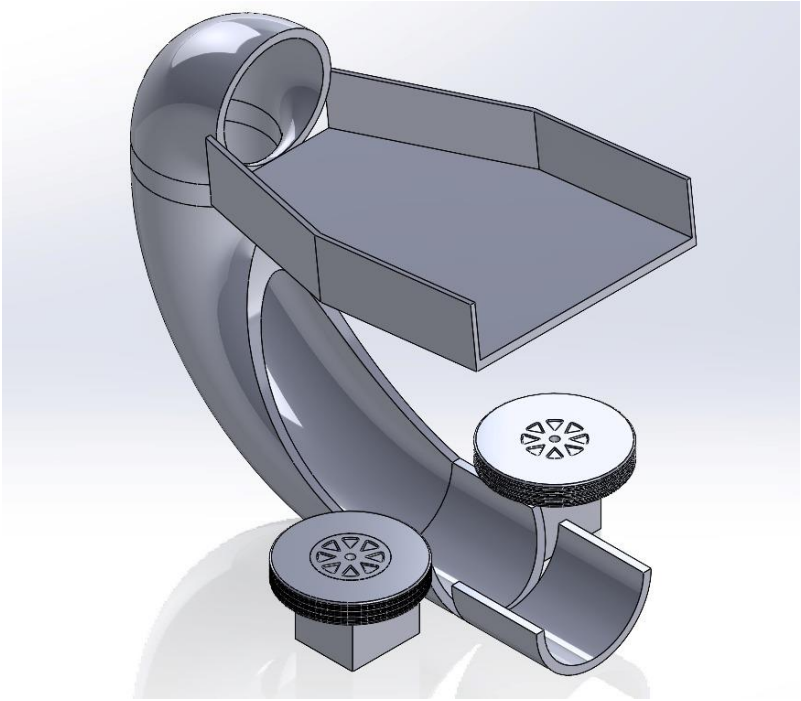


Figure 6 CAD model of initial design

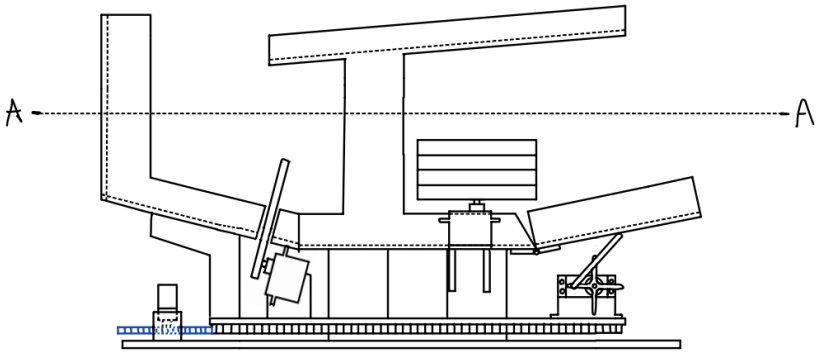


Figure 7 Hand-drawing of initial design

As the team has decided to utilize a counter-rotating wheels design, a preliminary design as shown in the figures above has been developed. The reservoir-shaped part will serve as a storage space for the balls, which will be tilted at an angle to allow for natural rolling down to a guided slope. At the end of the slope, a stopper controlled by a servo motor will be employed to halt the ball's movement. Furthermore, the angle of shooting can be adjusted by changing the tip angle of the pipe using another servo motor.

However, several problems were encountered during the design phase:

1. Compact Design: Achieving a compact design that adheres to the competition's size constraints has proven to be quite challenging.
2. Angle Adjustment: Adjusting the shooting angle by modifying the tip angle of the pipe resulted in the ball bouncing off instead of being effectively guided. This issue arises from the horizontal orientation of the wheels.
3. Misalignment and Ball Rotation: The misalignment between the tip angle and the rotation of the wheels may induce unintended rotation of the ball. To ensure precise and consistent shots, the team aims to address this misalignment issue and minimize any undesired ball rotation.
4. Manufacturing Challenges: The design presents manufacturing challenges due to the complexity of components such as the angled reservoir, guided slope, and servo motor-controlled stopper.

6.2 Second Design

Some problems in the initial design were identified and adjustments were made accordingly, as well as coming up with alternative designs. The major updated features and ideas are listed below:

PVC launching pipe

For the simpleness of structure and convenience of production, the curvy ball passage shown in the initial design was replaced with a PVC pipe that was cut in half. The PVC pipe was purchased at an offline shop and then processed so that grooves are preserved for the counter-rotating wheels to contact the ball.

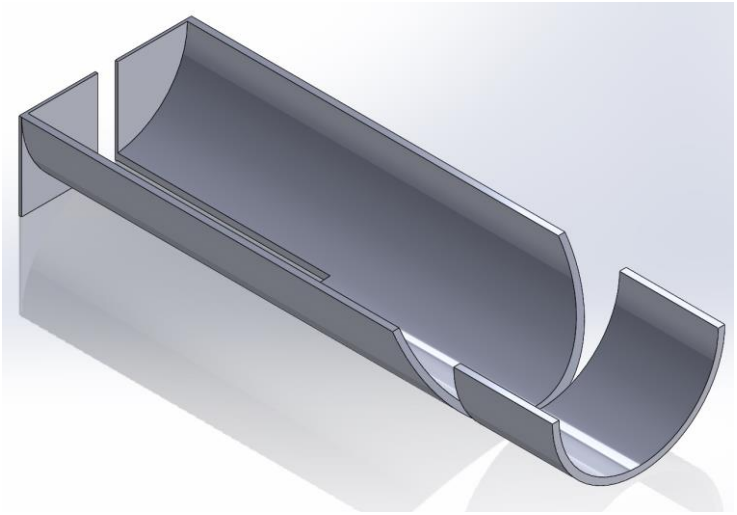


Figure 8 Launching passage

Ball pusher

The trap door added in the initial design was found unnecessary for non-stop shooting as it could disturb the flow of the ball. Starting from the second design, a leg-shaped ball pusher at the back side of the PVC pipe was adopted to push the ball smoothly. Further details will be provided in the latter (Final Design) part.

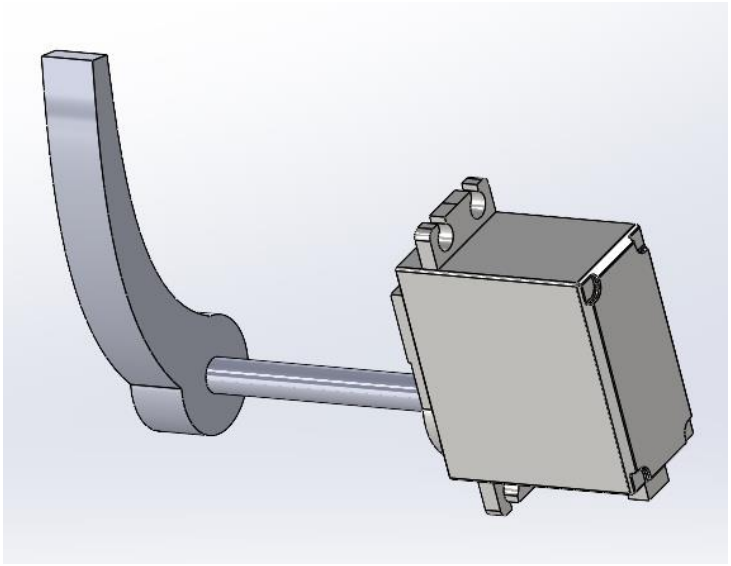


Figure 9 Ball pusher

Magazine clip and storage basket dismissed

These features needed excessive materials, space, and raised overall machine weight, making it harder for the motors at the bottom to work smoothly. The balls were loaded up one by one manually.

Angle launching methods

In our second design, the angle changing mechanism was roughly still the same as the initial design, which featured a separate plate in front of the horizontal launching pipe. However, to launch a ball with a projectile motion, a drastic angle change was needed at the interface of the horizontal launching pipe and the separate plate in front of it. In this case, the ball might bounce back upon hitting the separate plate and an unstable movement might occur.

To pursue better stability and consistency when launching the ball at a steeper angle, a new design was proposed, in which the whole device was set at an angle. As the whole device is tilted, the angle change between the pipe and the adjustable separate plane in front of it was not big, which reduced potential bouncing of the ball.

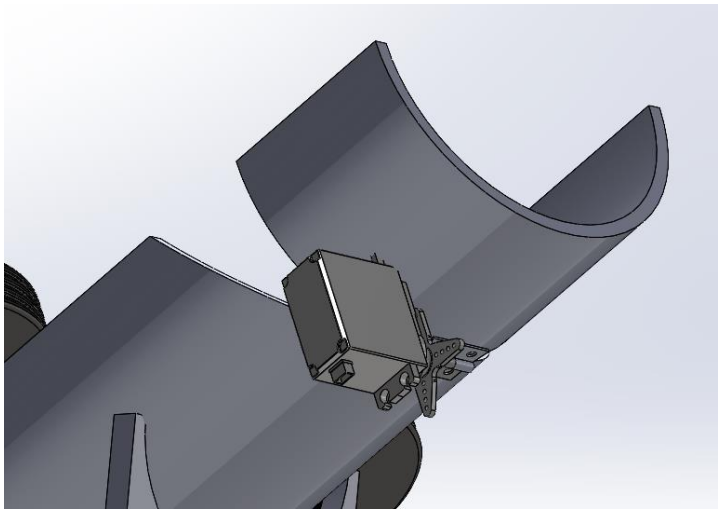


Figure 10 Angle-adjustable separate section

Mounting methods

The motors, including two DC motors for the counter-rotating wheels, a servo motor for the

adjustable separate plate in front of the launching pipe, and the other servo motor for the ball pusher, are all mounted on the PVC launching pipe. These motors are fixed to the pipe by motor stands that were to be designed later. The launching pipe, withstanding the weight of all the motors, was directly connected to two supports standing on the gear system at the bottom.

Other ideas

When an unspun ball travels through the air in a straight line, it can be affected by external forces, causing deviations from its intended path. To address this instability, the concept of imparting spin to the ball emerged.

The idea of spinning the ball gave rise to two potential designs involving rotating wheels, both aiming to give the ball a backspin, like shooting a basketball. The first idea involved placing wheels vertically and creating a speed difference between them. However, due to height restrictions on our shooting machine, this idea was not practical.

The other idea was to have a single spinning wheel at the bottom of the ball and add an upper cover to control the ball. However, predicting the ball's movement when accelerated by the wheel and hitting the upper cover proved challenging. As a result, these ideas were abandoned, and the original mechanism of two counter-rotating wheels at the same horizontal level was retained.

Also, the variation in size among the balls posed a potential factor that contributed to the inconsistent movement of the balls, as it influenced the friction between the wheels and the ball. To address this issue, it was preferable to establish a specific order for the balls within the three baskets.

Overall mechanism

The ball was placed at the back side of the PVC pipe manually. Upon being pushed by the ball pusher, the ball moved forward and was accelerated by the counter-rotating wheels. As the ball travelled through the pipe, the separate plate guided it to take off at a specific angle. The initial direction of the ball was determined by a pair of gears at the bottom and the slope of the separate plate in front of the pipe controlled by a servo motor as mentioned previously.

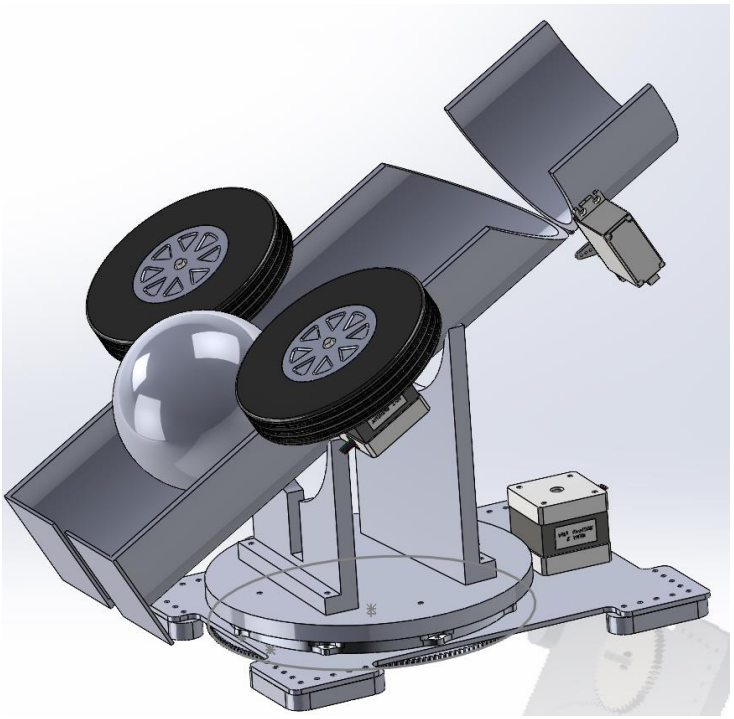


Figure 11 CAD model of second design

6.3 Final Design

Due to limitations of the previous designs, several changes and adjustments have been made throughout the period and below are the finalized design and features:

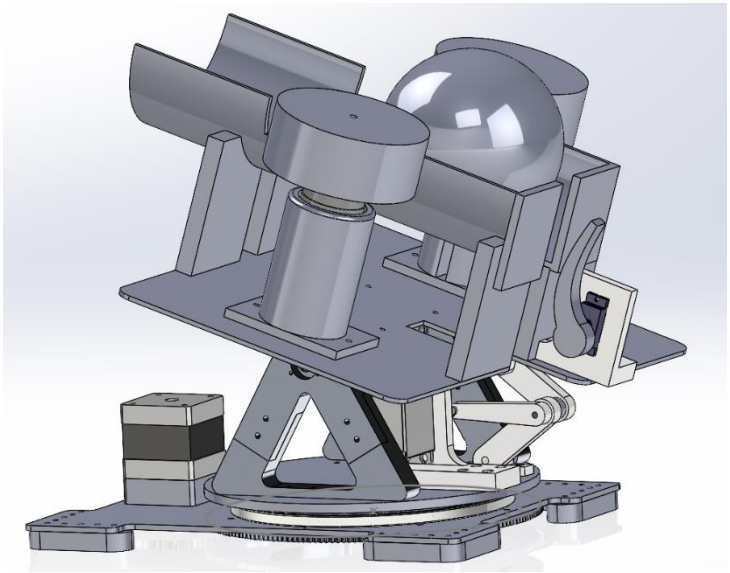


Figure 12 CAD model of final design

Counter-rotating wheels for shooting

As mentioned previously in the *'Research on Different Mechanisms'* part, counter-rotating wheels have high speed, consistency, and efficiency. Two DC motors with wheels of diameter 84mm are utilized for this purpose, mounted on two sides of the PVC pipe.

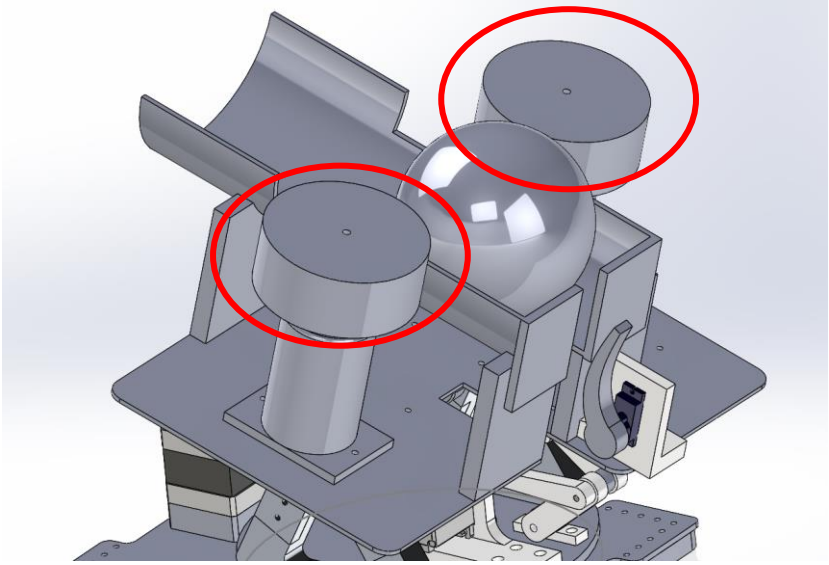


Figure 13 CAD model of counter-rotating wheels

Gear system for horizontal angle changes

The bottom board features a gear system for turning the upper plate connected to the launching system to control the horizontal angle changes. The stepper motor turns the smaller gear (spur gear of 56 teeth) connected by a D-shaped hole. The smaller gear is interlocked with a bigger gear (spur gear of 180 teeth), which is connected to the bottom support plate and inner bearing retainer. These are connected to the upper plate for turning the overall launching system.

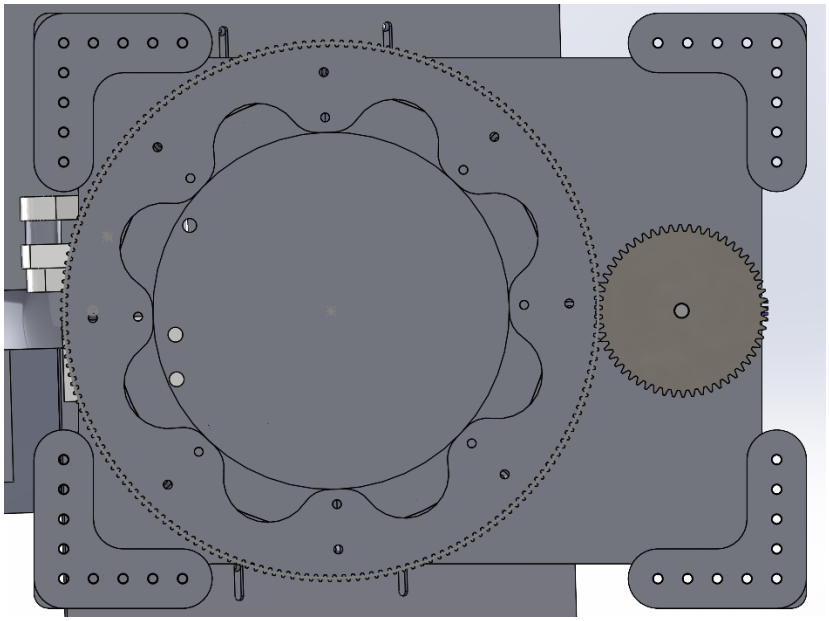


Figure 14 CAD model of bottom board with interlocked gears

In between the outer and inner bearing retainers, there should be a bearing for smooth rotation. However, as this is a one-time project and as the stepper motor can supply enough force to rotate the gears without a bearing if the friction between the board and gears can be lessened, for example, by using grease, another method is used eventually instead of the bearing, which will be explained further in the *'Manufacture'* part.

Linkage system for vertical angle changes

The idea of adding a separate plate in front of the launching pipe is not practical with various potential problems, and thus new systems for controlling vertical angles are implemented. Originally, using another gear system for the connection part between the PVC pipe and the bottom board was considered. However, if the gears are too small, they would not be able to stand the weight of the pipe, motors, mounting cases, and mounting board. On the other hand, if the gears are too big, the launcher would go over the height restraint. Therefore, the linkage system is implemented.

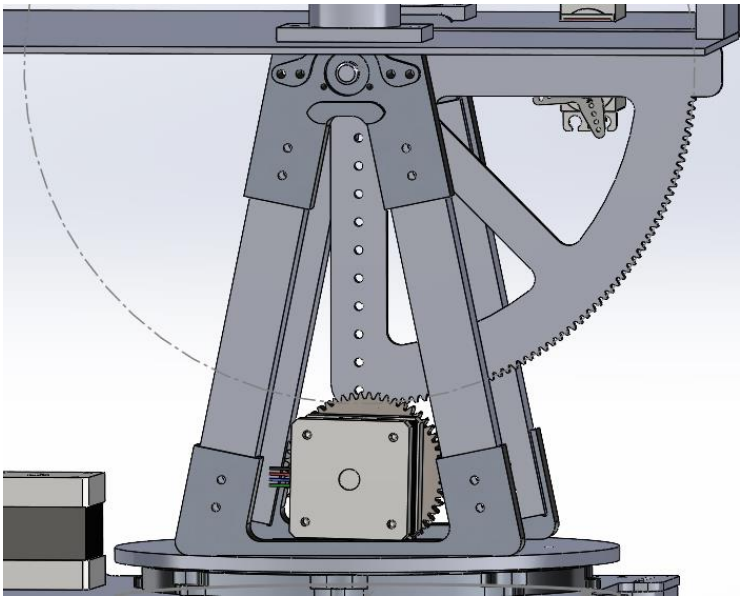


Figure 15 CAD model of gear system between bottom and upper parts

The linkage system features three linkages, each connecting to the upper plate of the bottom board, servo motor, and bottom mounting plate for the pipe and motors. The linkage system is connected to the bottom mounting plate, but the plate is connected to bearings with a rotating shaft, which allows smooth rotation for angle adjustments. Within 90 degrees of turning of the servo motor, the launcher can achieve the desired angle for launching the ball.

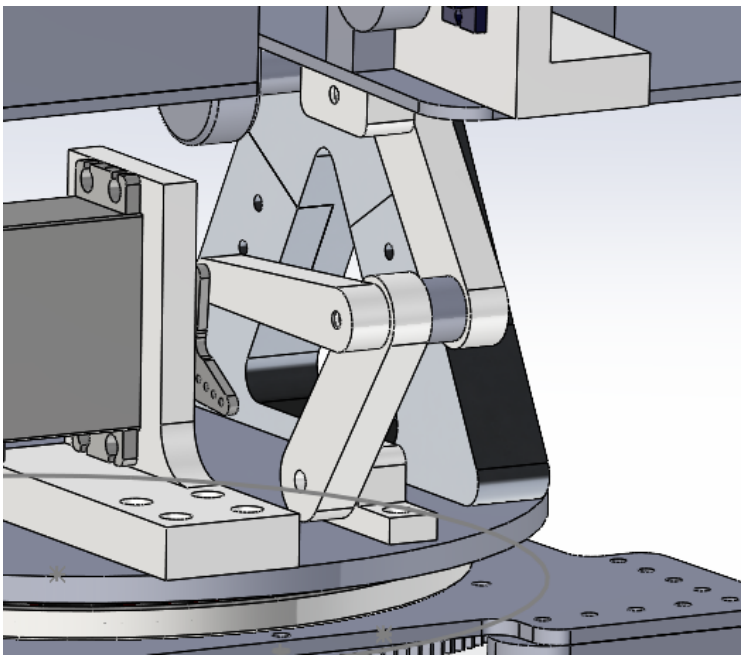


Figure 16 CAD model of linkage system between bottom and upper parts

PVC pipe and motors mounting board

Connecting the PVC pipe directly to the stand restricts vertical angle changes, making it necessary to add in a separate system to achieve the purpose, and adds difficulty in mounting the motors connected to the pipe and designing the appropriate stand. A separate plane is added beneath the pipe to achieve such purpose, holding the motors and the pipe firmly in place, making it easier to control the vertical angle at the same time.

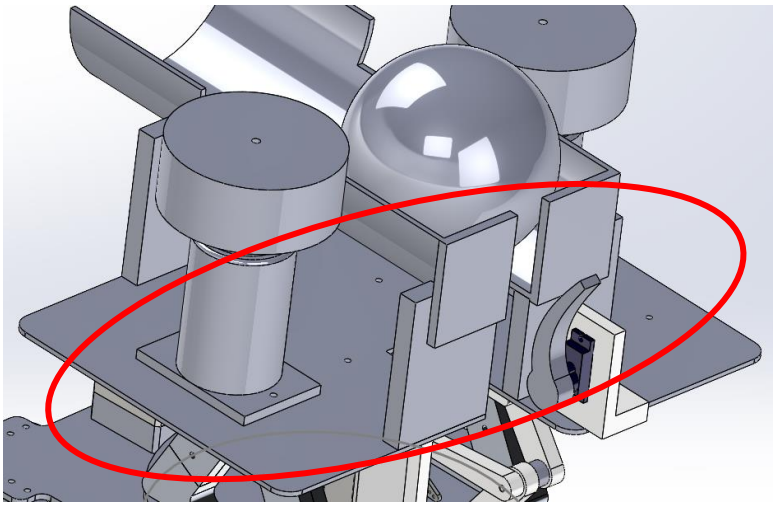


Figure 17 CAD model of PVC pipe and mounting board

Motor mounting stands

Different motor mounting stands are designed for stepper motors, servo motors of different sizes, and DC motors to ensure the motors stay in position when motors are in action. Small tolerances have been added in to compensate for the imperfectness of 3D printing and holes have been added to fix the motors with screws.

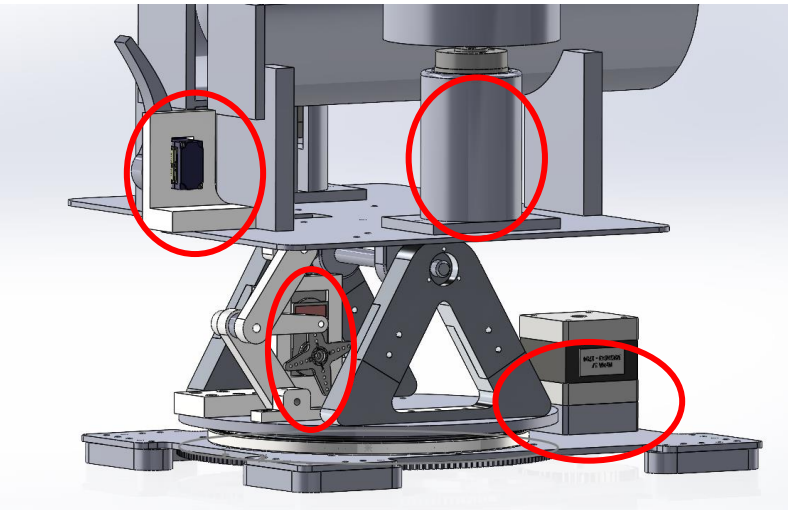


Figure 18 CAD model of motor mounting stands

Ball pusher

As mentioned previously in the 'Design & Problem Solving: Second Design' part, a ball pusher connected to a small servo motor located at the back side of the PVC pipe has been added to the design, ensuring constant starting speed and direction of the ball for the accuracy of the trajectile.

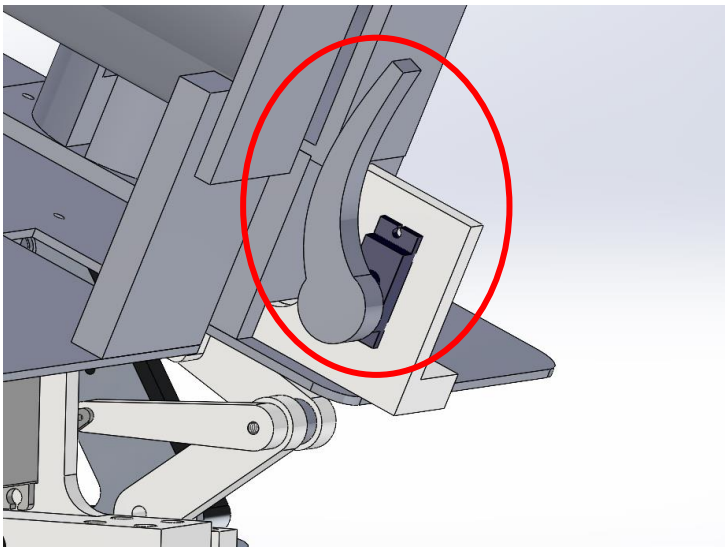


Figure 19 CAD model of ball pusher

7. Electronics

The launcher model incorporates various electronic components, including an Arduino board, a shell, switch buttons, a joystick, and motors (including DC motors, servo motors, and a stepper motor). Initially, the team members focused on individually controlling each motor to familiarize themselves with Arduino programming and understand the functionality of each programming code based on the motor type.

7.1 Use of Switch Buttons

Subsequently, connections were established between the switch buttons and motors to test controlling motors separately using four switch buttons respectively. By integrating all the relevant codes, specific functions were assigned to each switch button. For instance, the red button was assigned to halt the operation of the DC motor in case of an emergency. The blue button controlled the movement of the first servo motor, which operated the ball pusher in a back-and-forth motion. The yellow button regulated the stepper motor, enabling adjustments to the horizontal angle by rotating the model's base. Lastly, the green button governed the second servo motor, facilitating adjustments to the vertical shooting angle. Nonetheless, the use of switch buttons was discarded in the later stage as all necessary controls were able to be performed with only the joystick.

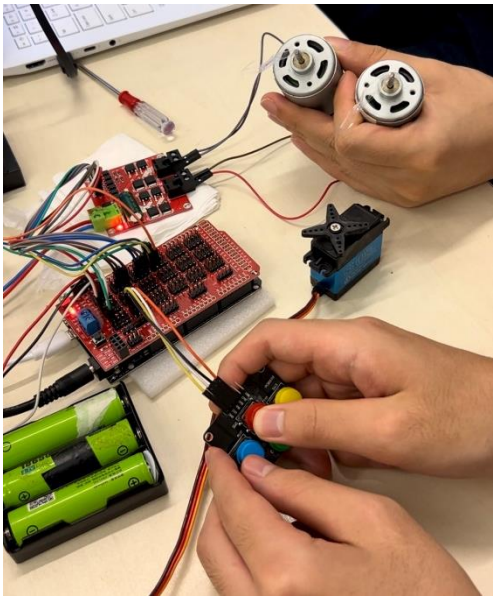


Figure 20 Switch buttons to control motors

7.2 Use of Joystick

In addition, implementation of a joystick to provide manual control over both servo and stepper motor were experimented. The purpose was to allow for manual adjustment of the shooting angle by hand, particularly when the initially set angle did not result in successful shots. This feature enabled real-time modifications to be made during the shooting process, ensuring optimal accuracy and performance.

- Left joystick x-direction for stepper (horizontal shooting angle change)
- Left joystick y-direction for servo motor (vertical shooting angle change)
- Left joystick k button for servo motor (90-degrees back-and-forth motion of pusher)
- Right joystick k button for automation (pre-set positions, position 1 with the first press, position 2 with the second press, position 3 with the third press)

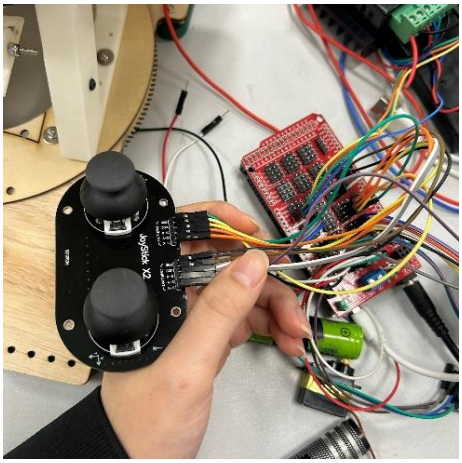


Figure 21 Joystick connected to Arduino board

7.3 Use of Receiver and Transmitter

Due to technical difficulties in controlling the DC motors more precisely alongside other motors, the combination of receiver and transmitter was used to control multiple motors simultaneously. Moreover, separate control allowed more team members to be engaged in the control of the launcher and from a distance with wireless control.

1. Receiver

   The receiver receives signals transmitted by the transmitter and relays them to the appropriate components, enabling remote control and precise command execution. For selecting the appropriate receiver, the receiver was checked if it was compatible with the transmitter.



Figure 22 Receiver R8

2. Transmitter

   The transmitter transmits control signals wirelessly to the receiver. Equipped with joysticks, switches, and knobs, the transmitter provides control over the connected motor functions and responses.



Figure 23 Transmitter

7.4 Technical Difficulties

As the experiment progressed, technical difficulties concerning the motor driver and stepper motor were encountered.

Troubleshooting Motor driver issue

The motor driver was found to be physically damaged internally, necessitating its replacement with a functional unit. Moreover, upon identifying the damage caused by a short circuit, a new driver board was installed for the DC motor and the Arduino board.

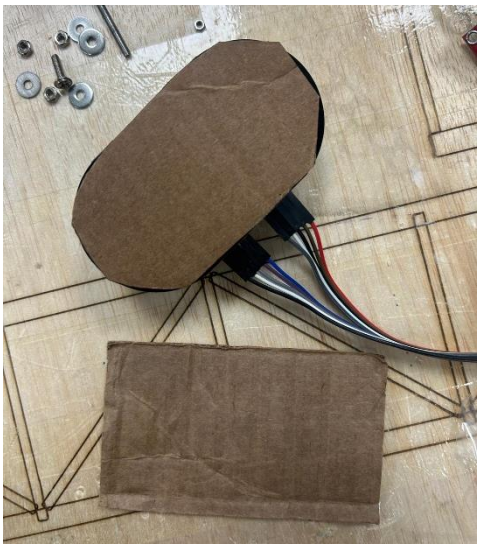


Figure 22. Board at the bottom of DC motor and Arduino board

Troubleshooting Stepper motor issue

Regarding the stepper motor, a problem with an insufficient voltage supply was identified. The existing three-cell battery, rated at 12V, proved inadequate to power the stepper motor effectively. Seeking guidance from a technician, a higher voltage power supply was utilized. This new power supply allowed us to adjust the maximum voltage to 30V, enabling the stepper motor to operate reliably.

Despite supplying the stepper motor with higher voltage, it was discovered that the inadequate rotation of the stepper motor was caused by two other factors: incorrect interpretation of the micro step driver setting and improper connection between the stepper motor and the stepper motor driver. To rectify these issues, the micro-step resolution was adjusted to 32 and the number of steps per revolution to 6400, according to the correct method of reading the stepper motor driver's micro step setting. Additionally, necessary changes to the wire connections were made, ensuring proper alignment between the stepper motor and its driver.

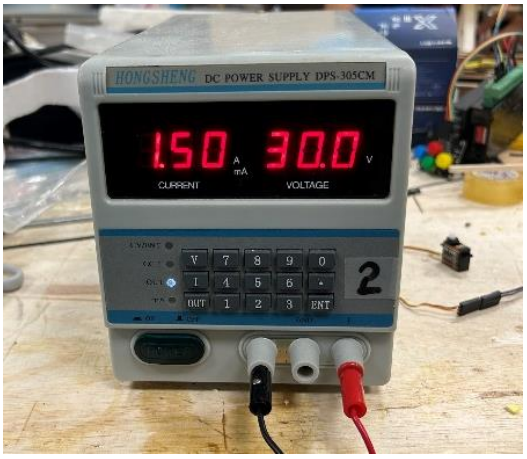


Figure 24 DC power supply with maximum of 30V

8. Manufacture

8.1 Fabrication

For manufacturing the launcher, major components with curvatures or complex shapes, including motor mounters, connection stands, linkage system, and bottom board stands were 3D printed. Other components, including bottom board, upper board, gear connection boards, and gears were laser cut where plywood was chosen as the main material for its strength.



Figure 25 3D printed parts



Figure 26 Laser cut parts

Other manual works were also conducted, such as sawing the PVC pipe for the launching passage, sanding the cut surfaces of the PVC pipe and 3D printed parts for tolerance, drilling holes at necessary parts, gluing different components together, and screwing screws and nuts.



Figure 27 Sawing PVC pipe



Figure 28 Gluing using epoxy glue

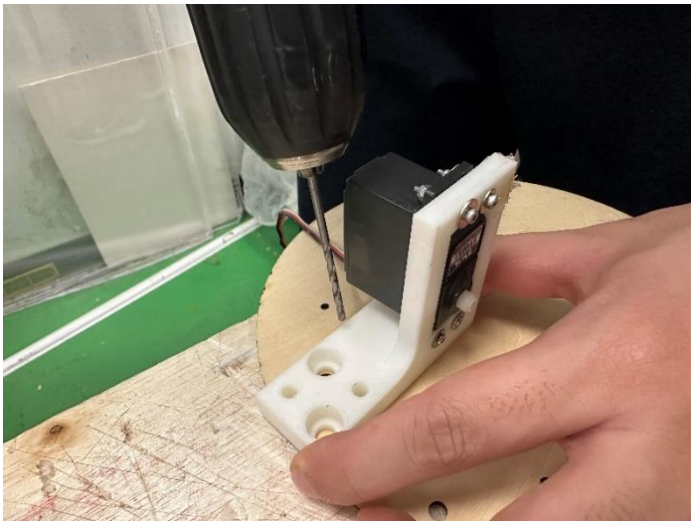


Figure 29 Drilling on 3D printed components

8.2 Problems Encountered

Gears connected to the stepper motor (small gear) and the entire system (big gear) were laser cut out of plywood as well as suggested from previous consultations that the entire system is going to be used for only a short period of time and no need for the gears to be cut out from stronger pieces like metal. However, as all components above the bottom plate were too heavy, with just 2 DC motors and pipe weighing around 800g, along with the problem of friction that is to be explained in the following paragraph, plywood gear teeth were not strong enough to rotate such heavy system and were worn out. As a result, the gears were laser cut again with thicker acrylic plates instead of plywood as a replacement.



Figure 30 Damaged gear teeth



Figure 31 Acrylic gears



Figure 32 Weight of DC motors and PVC pipe

Moreover, it was also suggested from previous consultations to use lubricants such as grease for the bottom gear-connected rotating part instead of buying a big bearing, once again, for the reason of short-time usage. Nonetheless, as all components above the bottom plate were too heavy, much friction was created between the bottom board and the gear, damaging the plywood gear teeth and the connection between the stepper motor and the gear. Therefore, Vaseline, instead of grease, which is not a feasible option for wooden materials, was used to minimize the friction and make the surfaces as smooth as possible.



Figure 33 Vaseline as lubricant

There was also the problem of difference between SolidWorks drawings and actual manufactured components. The holes on laser cut boards were slightly bigger and those on the 3D printed parts were slightly smaller than those on SolidWorks drawings. Also, the bottom board was manufactured thicker than the one in SolidWorks file as two layers of 3mm plywood was glued together since one layer was not strong enough to withstand all the components. As a result, the shaft of the stepper motor was too short when installed on the board, so the shaft-gear connection was elongated with more support so that the gear can be placed at desired location. Lastly, minor differences of manufactured parts led to geometrical deviation in stroke of the linkage system from the simulation in SolidWorks file.

8.3 Final Assembly

Below is the final assembly after all the necessary amendments during the manufacturing process:

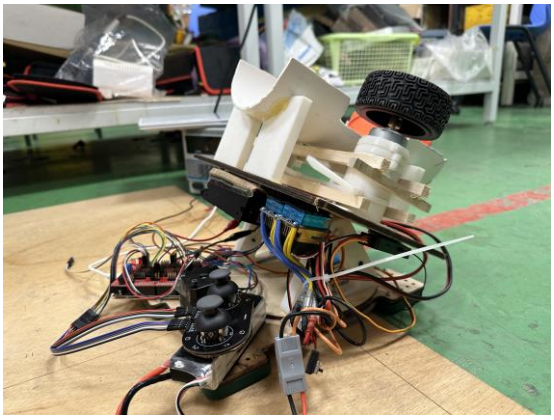


Figure 34 Final assembly

9. Testing

Automation to manual control

The initial experimentation with automation for the application revealed its unsuitability due to the inconsistent shape of the ball, leading to unpredictable shooting power. Moreover, the implementation of a powerful DC motor resulted in excessive vibrations throughout the board. Consequently, for the real-life competition, the team discontinued the automation approach and instead adopted a 'Try until success' strategy to accomplish the goal of shooting all three baskets in the shortest possible time. This approach entails manually adjusting the shooting angle for each trial, resulting in improved control, and shooting accuracy.

Vibration of DC motor

Due to the significant strength of the vibrations generated by the DC motor, measures were taken to address this issue in the project. The team recognized the necessity of utilizing cable ties to ensure that the motor mounters remained securely fastened, given the strong vibrations produced by the DC motors. To further mitigate the impact of vibrations, wooden sticks were introduced on the side of DC motor mounters on the upper board. This addition aimed to minimize vibrations and maintain stability within the components. Additionally, the upper board was replaced with a thicker plywood board to enhance overall structural integrity and withstand the vibrations more effectively.

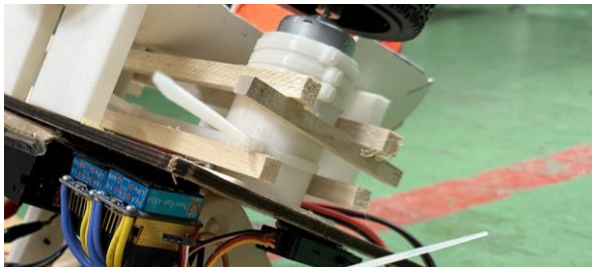


Figure 35 Wooden sticks on motor mounts for reducing vibration

Separate control for DC motors

As mentioned in the previous section, due to difficulty in controlling DC motors alongside other motors including servo and stepper motors, a separate control was used for the control of DC motors using a receiver and a transmitter. This allowed more precise speed control of

the motors. The motors were connected to the electronic speed controller (ESC), which was then connected to the receiver to receive signals from the transmitter (radio controller). Thicker wires were used to connect the motors to ESC.

10. Evaluation

This project has provided team members with a valuable opportunity to engage in the complete process of designing a shooting machine using an engineering approach. This encompassed problem identification, brainstorming solutions, exploring alternatives, developing a design proposal, creating a prototype, conducting rigorous testing and evaluations, and iteratively refining the design. Throughout this project journey, every team member has gained profound insights from two significant lessons.

Allocating Sufficient Time for Prototype Testing and Refinement

One crucial lesson our team learned was the importance of allocating ample time for thorough testing and refinement of the prototype. However, we encountered challenges due to the timing of our initial prototype. A substantial portion of our time was spent revising the SolidWorks design, resulting in unforeseen issues during the subsequent production and assembly stages.

On the hardware side, we faced difficulties in accurately estimating the impact of the weight factor during the design phase. The components above the bottom plate turned out to be heavier than anticipated, leading to excessive friction between the bottom board and the gear. Consequently, the plywood gear teeth were damaged, and the connection between the stepper motor and the gear was compromised. Additionally, the high-speed rotation of DC motors caused vibrations that resulted in cracks on the board and linkages, as well as inconsistencies in the launch mechanism.

We also encountered minor manufacturing-related issues, such as laser-cut wood holes being larger than the original design and 3D printed parts having relatively smaller holes. This necessitated additional drilling during the manufacturing process. These problems, while not

directly related to the design, required modifications or re-manufacturing of certain parts, resulting in additional time consumption.

Regarding control, we faced more challenges than anticipated, including motor failures and short circuits leading to burnouts. Issues with power supply voltage, wiring, program adjustments, and even changes in control mode further extended the time required for our machine to operate smoothly after hardware assembly. Additionally, we needed extra time to fine-tune the launch elevation and wheel speed.

These experiences have taught us the critical lesson of allocating sufficient time for prototype adjustments after its completion, rather than assuming flawless functionality from the initial assembly. By allowing extra time, we can effectively address unforeseen issues, make necessary modifications, and ensure that the prototype operates as intended.

Avoiding Complex Mechanisms and Embracing Simplified Design

One of the primary challenges we encountered during this project revolved around controlling the horizontal and elevation angles. To address horizontal angle control, we utilized a stepper motor and a pair of gears. For elevation angle control, we employed a servo motor to drive connecting linkages. Although gears and linkages are common methods in the mechanical industry for force transfer, we faced difficulties arising from material and manufacturing limitations, which became the source of our problems.

For instance, gears made of plywood boards experienced tooth breakage under excessive friction, and 3D-printed linkages broke multiple times. The complexity of the connecting rod's geometry also added to the design and control challenges. Additionally, there were minor differences between the simulated results in the SolidWorks file and the manufactured parts, leading to geometrical deviations in the stroke of the linkage system.

Upon reviewing the other team's designs, we realized that employing a high-torque motor to connect all the shafts could create a simpler mechanism. This approach would help us avoid the difficulties encountered in this project. It prompted us to reflect on whether we could design simpler mechanisms without restricting our ideas based on what we learned in the

mechanical engineering department.

By exploring alternative approaches and breaking free from conventional design constraints, we can potentially develop innovative solutions that are not limited by the challenges faced in this project. This realization encourages us to think more creatively and consider simpler designs that can effectively achieve the desired objectives.

11. Conclusion

This project gave us invaluable insight into the design and development of a realistic electromechanical product by manufacturing a semi-automatic ball shooter. Along with applying conventional mechanical technology knowledge that we have acquired, it allowed us to gain hands-on experience in utilizing computer-aided methods, such as Arduino coding, to incorporate automatic control using a joystick and switch buttons into our product design and development processes.

In our project, our main goal was to achieve the fastest possible shooting of three baskets within a time limit of 5 minutes. To accomplish this, we implemented various components and mechanisms. A DC motor was utilized to provide the necessary kinetic energy to accelerate the ball towards the basket. The servo motor was employed to regulate the movement of the linkage system and ball pusher system. The stepper motor was responsible for controlling the movement of the gear system. For automation, we used a joystick to regulate the movements of these motors. To control the speed of the DC motor, we integrated an Electronic Speed Controller (ESC) with a Transmitter and Receiver system.

During the testing, automation was initially considered but proved unsuitable due to inconsistent ball shape and unpredictable shooting power. As a result, manual control was implemented, leading to improved accuracy. Vibration issues caused by the DC motor were addressed by securing motor mounters and introducing wooden sticks for stability. ESC was utilized for the DC motor control.

To conclude, the project provided valuable insights, highlighting the significance of allocating adequate time for prototype refinement. Simplifying design mechanisms and nurturing creativity proved to be effective strategies for promoting innovation. The project met our initial expectations and objectives, deepening our understanding of the intricacies of shooting machine development and paving the way for future advancements.

12. Appendix

Drawings

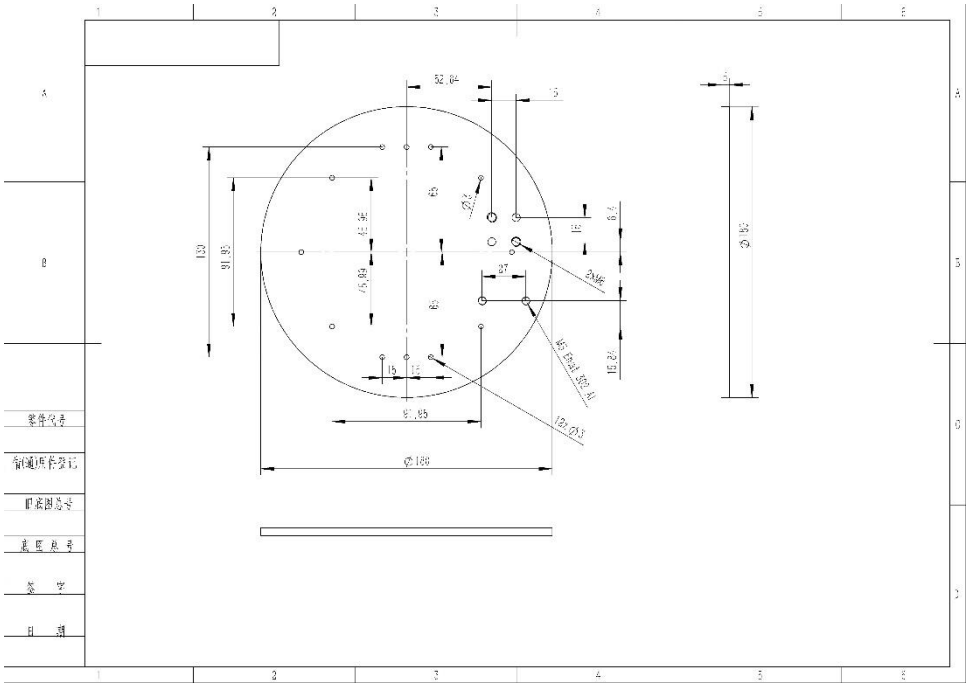


Figure 36 Drawing of rotating plate

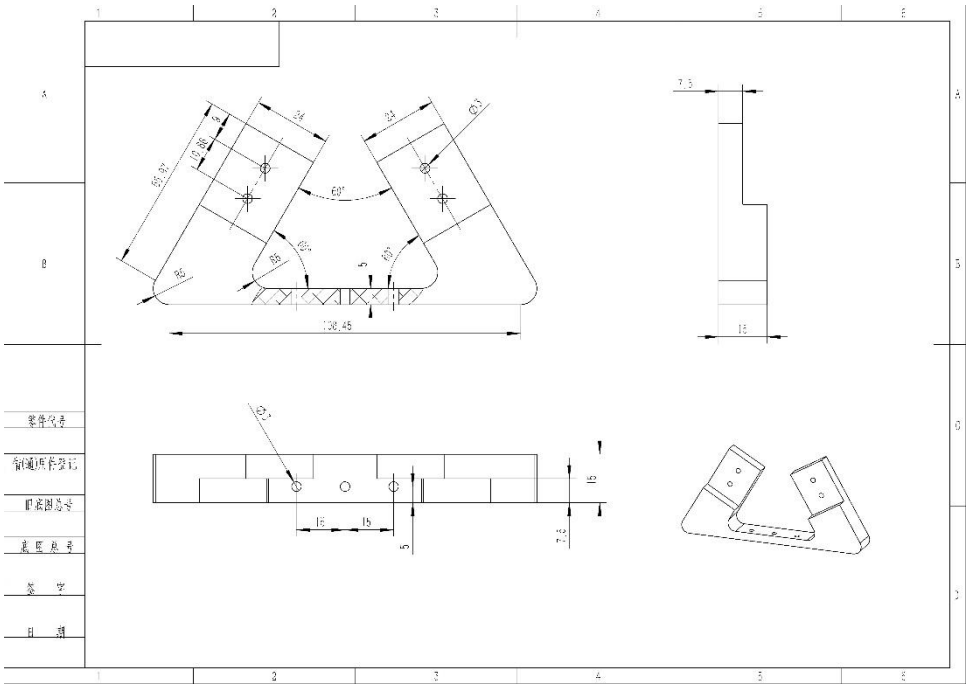


Figure 37 Drawing of main stand (lower part)

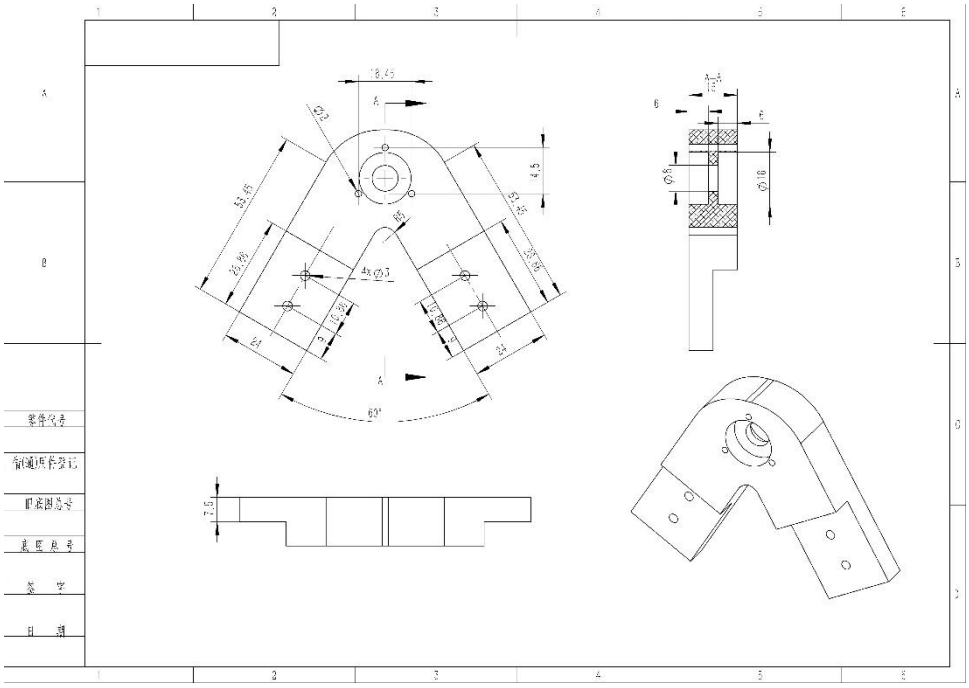


Figure 38 Drawing of main stand (upper part)

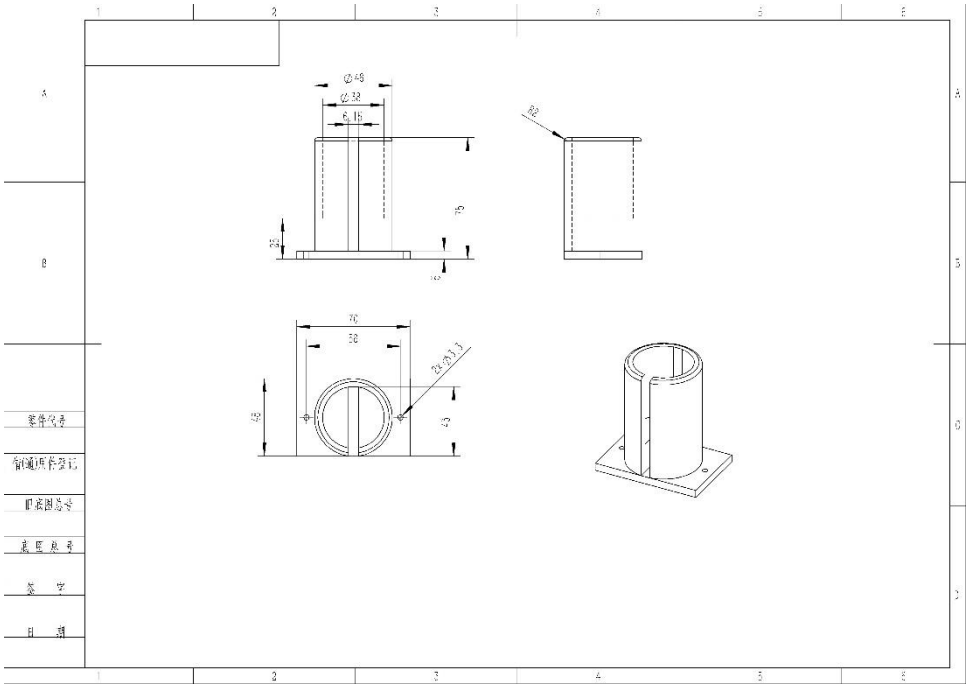


Figure 39 Drawing of DC motor mount

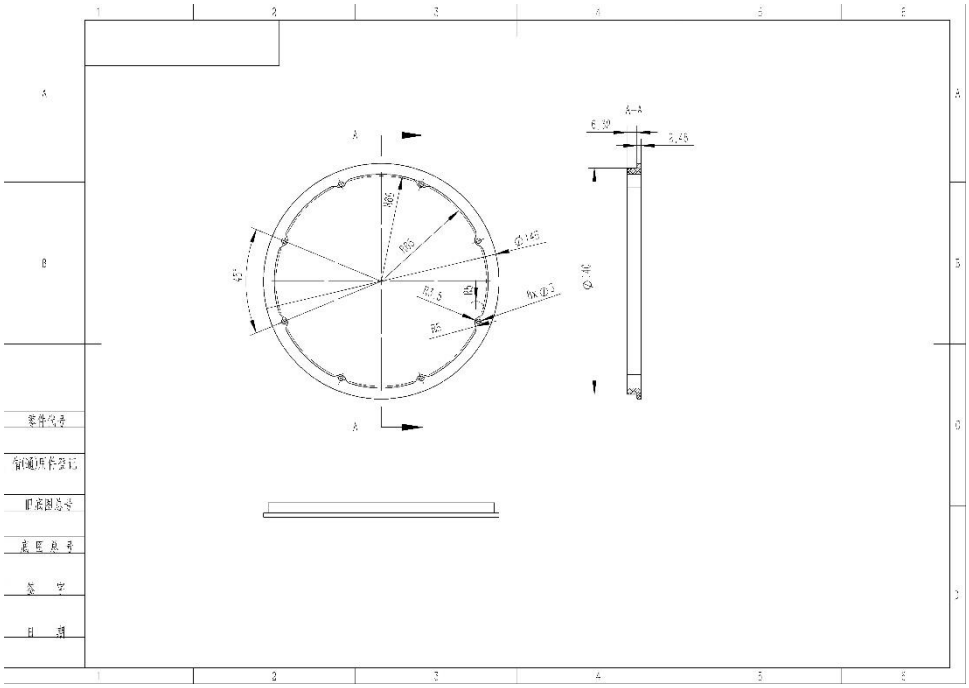


Figure 40 Drawing of bottom board bearing retainer

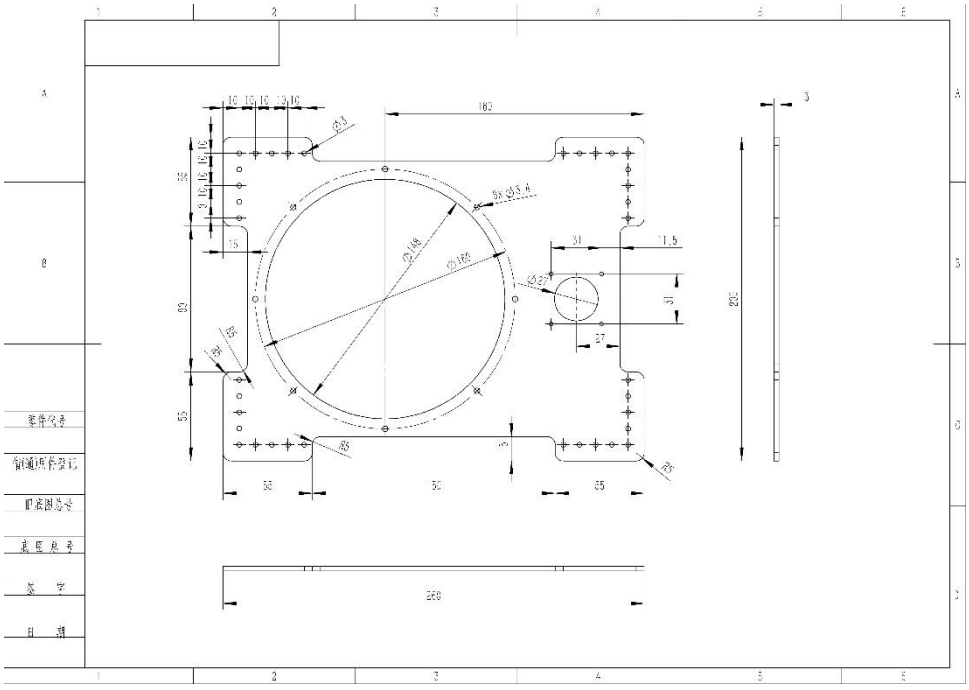


Figure 41 Drawing of bottom plate

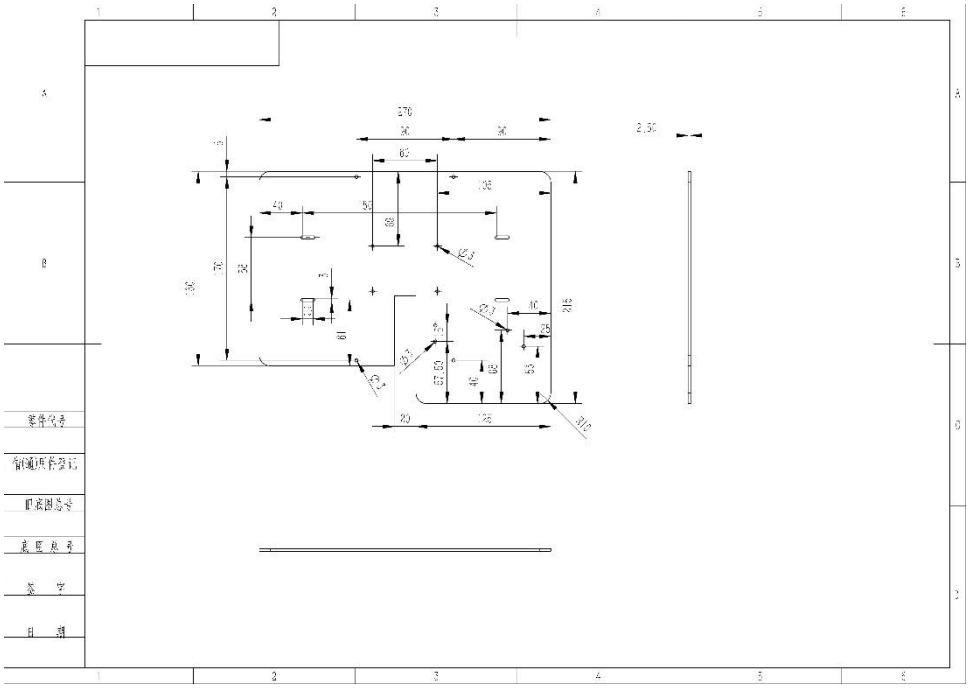


Figure 42 Drawing of upper plate

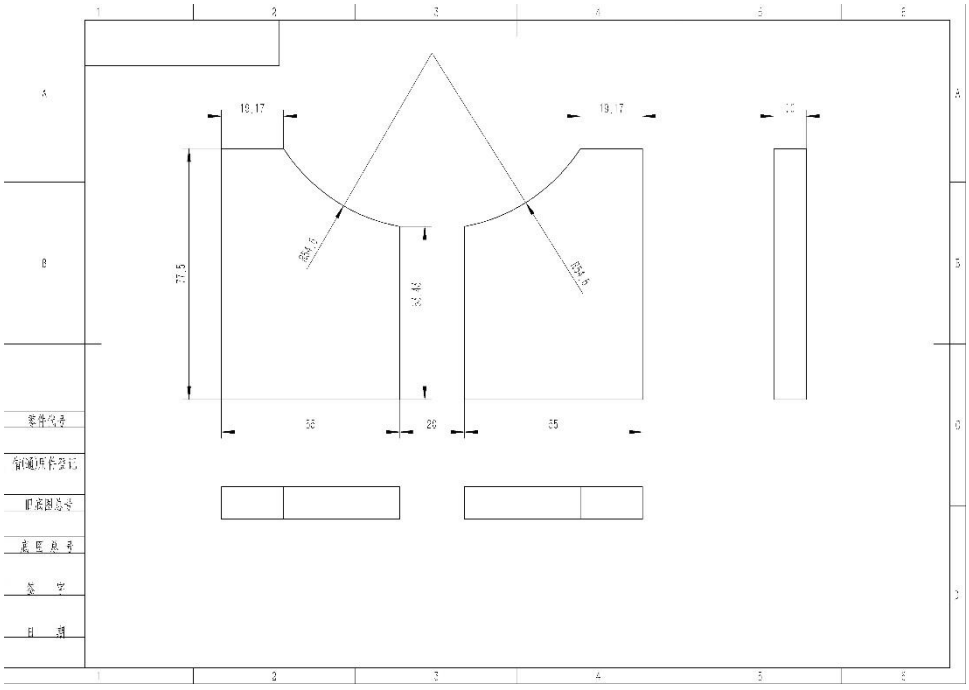


Figure 43 Drawing of launching track stand

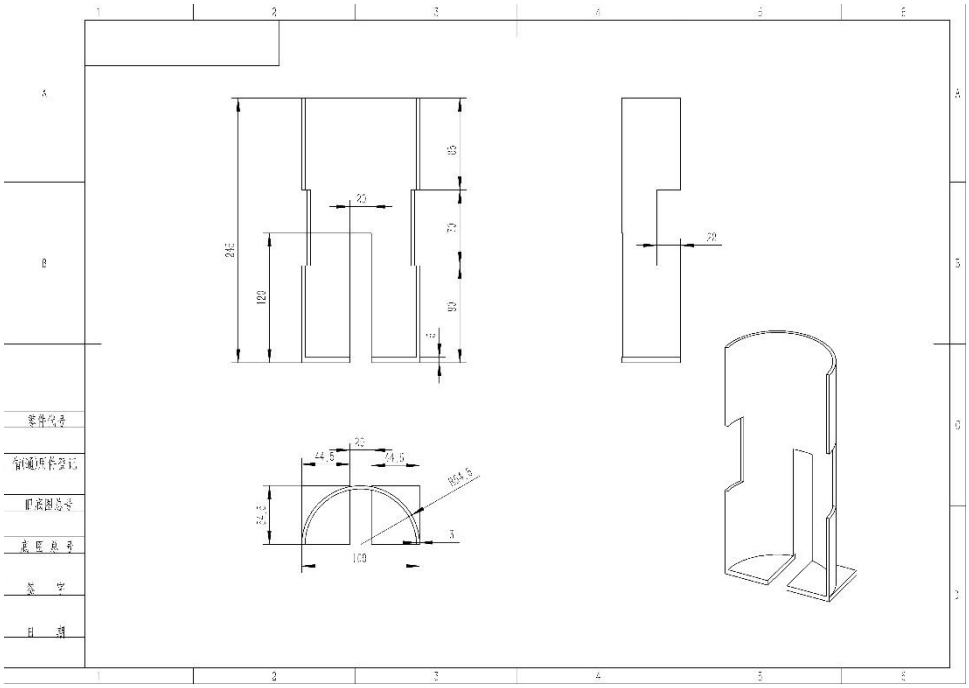


Figure 44 Drawing of launching track

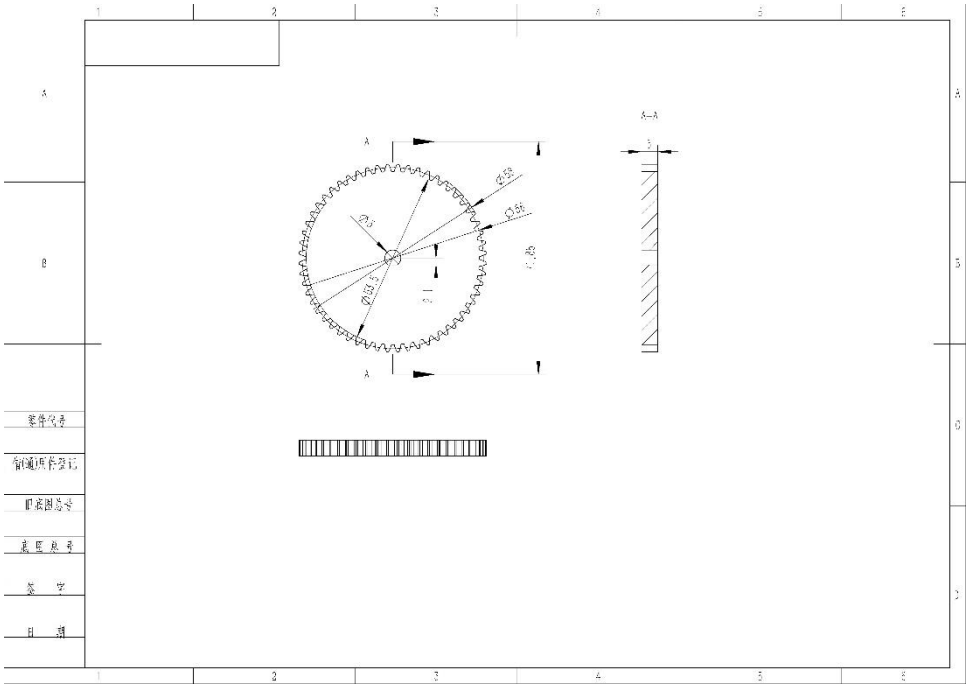


Figure 45 Drawing of smaller gear

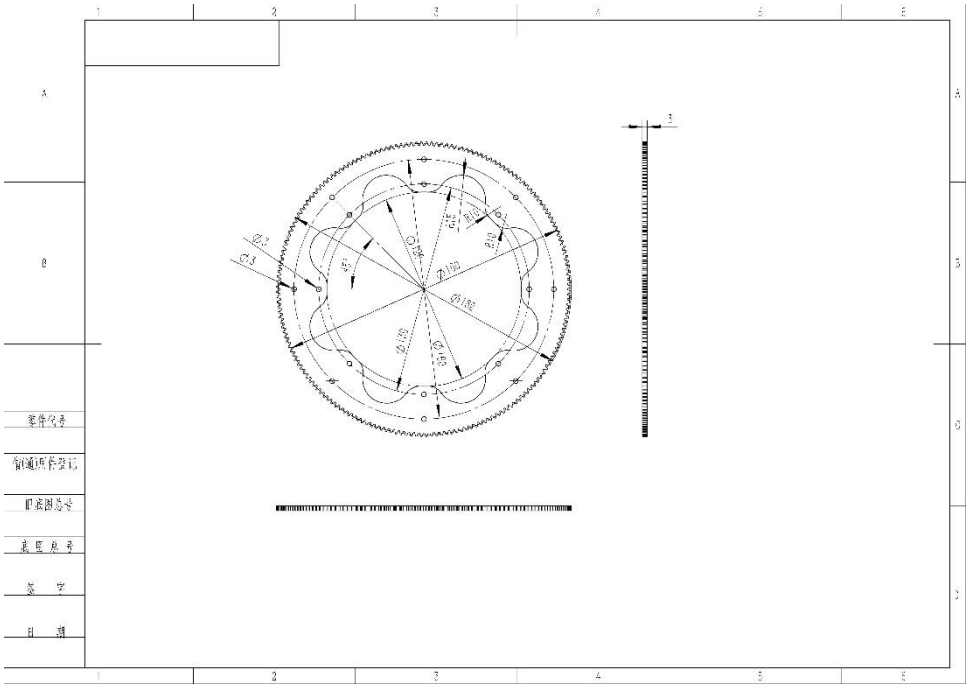


Figure 46 Drawing of bigger gear

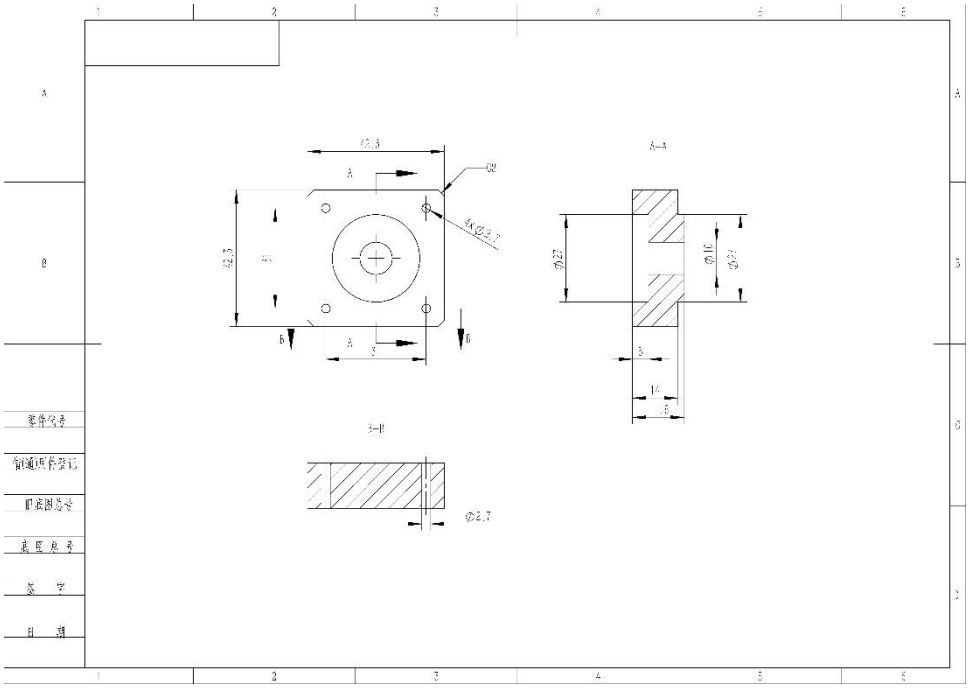


Figure 47 Drawing of stepper motor mount

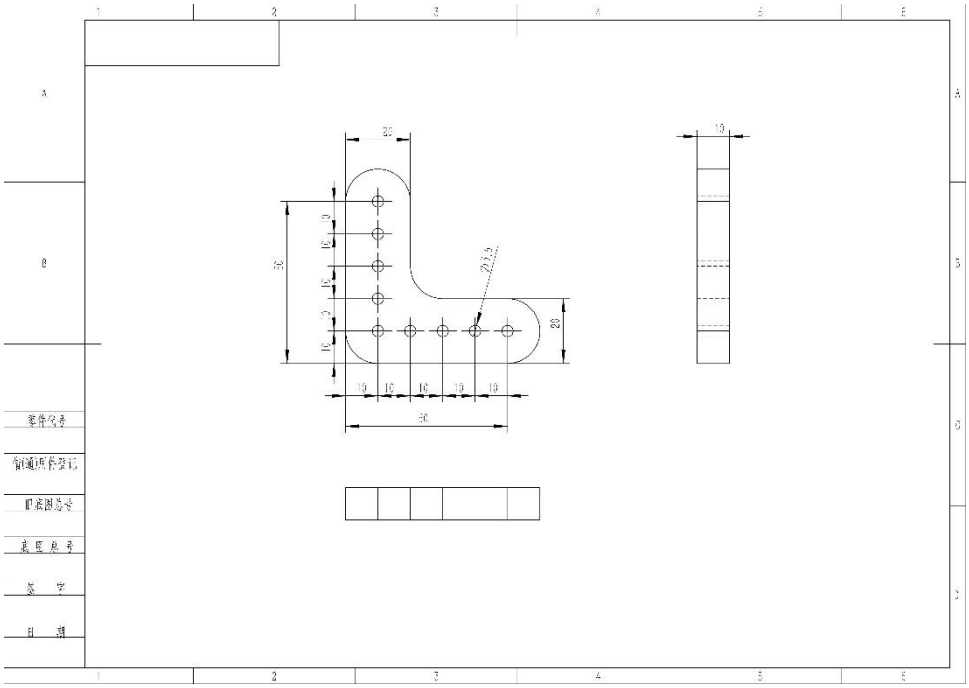


Figure 48 Drawing of bottom plate support

Codes

Joystick to control motors:

```
#include <Servo.h>
#include <Stepper.h>

Servo servo1;  // First servo on pin 12
Servo servo2;  // Second servo on pin 13
const int SERVO_PIN1 = 12;
const int SERVO_PIN2 = 13;
const int STEPPER_PIN1 = 2;
const int STEPPER_PIN2 = 3;
const int JOYSTICK_X_PIN = A0;
const int JOYSTICK_Y_PIN = A1;
const int JOYSTICK_K_PIN = A2;

const int STEPS_PER_REVOLUTION = 6400;  // Number of steps per revolution
const int MICROSTEP_RESOLUTION = 32;    // Microstep resolution
Stepper stepper(STEPS_PER_REVOLUTION * MICROSTEP_RESOLUTION, STEPPER_PIN1, STEPPER_PIN2);

const int JOYSTICK_THRESHOLD = 600;  // Adjust threshold value as needed

bool isDefaultDirection = false;     // Flag to track if joystick is in default direction
bool isSecondServoRotated = false;    // Flag to track the state of the second servo motor

enum ServoState {
```

```
  SERVO_IDLE,
  SERVO_ROTATING,
  SERVO_ROTATED
};

ServoState servo2State = SERVO_IDLE;  // Initial state of the second servo motor

void setup() {
  servo1.attach(SERVO_PIN1);
  servo2.attach(SERVO_PIN2);
  stepper.setSpeed(100);
}

void loop() {
  // Read the joystick X-axis value for the stepper motor control
  int joystickX = analogRead(JOYSTICK_X_PIN);

  // Check if joystick value exceeds the threshold or if it's in the default direction
  if (joystickX > JOYSTICK_THRESHOLD || joystickX < (JOYSTICK_THRESHOLD / 2)) {
    // Map the joystick X-axis value to the number of microsteps for stepper motor rotation
    int microsteps = map(joystickX, 0, 1023, -STEPS_PER_REVOLUTION * MICROSTEP_RESOLUTION,
STEPS_PER_REVOLUTION * MICROSTEP_RESOLUTION);

    // Rotate the stepper motor
    stepper.step(microsteps);

    // Update the default direction flag based on the joystick value
    if (joystickX > JOYSTICK_THRESHOLD) {
      isDefaultDirection = false;
    } else if (joystickX < (JOYSTICK_THRESHOLD / 2)) {
      isDefaultDirection = true;
    }
  }

  // Read the joystick Y-axis value for the first servo motor control
  int joystickY = analogRead(JOYSTICK_Y_PIN);

  // Read the joystick K direction value
  int joystickK = analogRead(JOYSTICK_K_PIN);

  // Map the joystick Y-axis value to servo angle range (0 to 180 degrees)
  int servoAngle1 = map(joystickY, 0, 1023, 0, 180);

  // Move the first servo to the calculated angle
  servo1.write(servoAngle1);

  // State machine for the second servo motor
  switch (servo2State) {
    case SERVO_IDLE:
```

```
      if (joystickK < (JOYSTICK_THRESHOLD / 2)) {
        servo2State = SERVO_ROTATING;
        isSecondServoRotated = !isSecondServoRotated;  // Toggle the rotation state
      }
      break;

    case SERVO_ROTATING:
      if (isSecondServoRotated) {
        servo2.write(90);
      } else {
        servo2.write(0);
      }
      servo2State = SERVO_ROTATED;
      break;

    case SERVO_ROTATED:
      if (joystickK >= (JOYSTICK_THRESHOLD / 2)) {
        servo2State = SERVO_IDLE;
      }
      break;
  }

  // Add a small delay between motor movements
  delay(10);
}
```